

4.7. Applications of Flip-Flops

Some of the common applications of flip-flops are as follows:

1. Switch bouncing.
2. Registers.
3. Counters.
4. Memory elements.

4.8. Race Around Condition

JK flip flop suffers from the problem of race around condition. When $J = 1$ & $K = 1$, is applied to the JK flip flop and JK flip flop is level triggered then output of the JK flip flop toggles so many times during the pulse width of the clock and output of the flip flop settled either at 1 or 0 depending upon the pulse width of the clock and propagation delay of the flip flop is called race around condition.

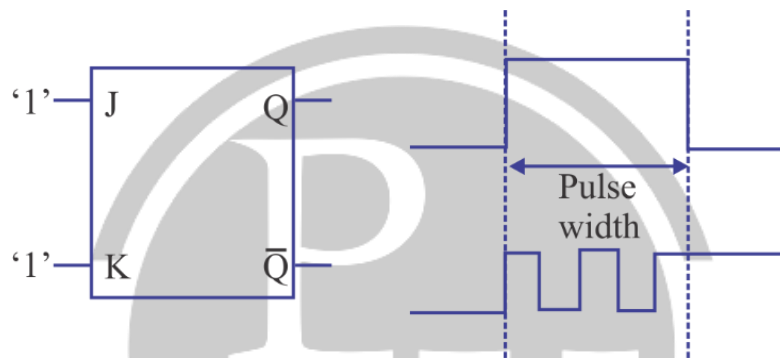


Fig. 4.21. Race Around Condition in JK Flip Flop

To avoid Race Around condition:

- $T_{\text{pulse-width}} < T_{\text{pd}} < T_{\text{clock}}$
- Master Slave flip flop

4.8.1. Master Slave Flip flop:

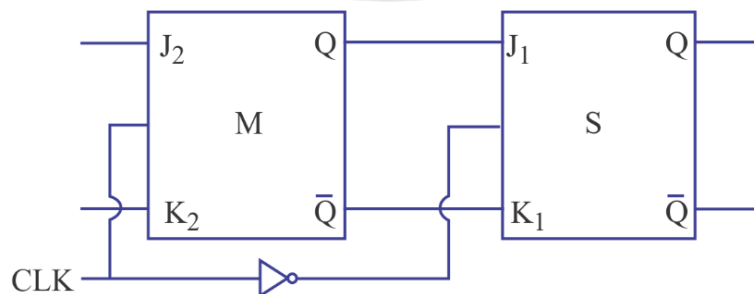


Fig. 4.22. Logic Diagram for Master Slave JK Flip Flop

- (a) In master slave flip flop, inverted clock is given to the slave.
- (b) Master slave flip flop is used to store single bit because output is taken only from slave flip flop.
- (c) Here, master flip flop is level triggered while slave is negative edged triggered.

Note: JK flip flop is also known as Universal flip flop.

4.9. Designing of One Flip Flop by Other Flip Flop

The steps for designing of one flip flop or new flip flop using existing or same existing flip flop.

Step 1: Write the characteristic table for the designed flip flop.

Step 2: Write the excitation table for the available flip flop.

Step 3: Write the logical expression.

Step 4: Minimize the logical expression.

Step 5: Circuit Implementation.

4.10. Shift Registers

An array of flip-flops is required to store binary information, and the number of flip-flops required is equal to the number of bits used to store is referred as registers.

Examples of registers are general purpose registers, flags, etc.

Now, the information or data can be stored or entered in serial form (one-bit at a time) or in parallel form (all the bits simultaneously) and can be retrieved like this manner too. The data will be entered or retrieved in serial form is known as temporal code and which is in parallel form is called special code.

Hence, registers can be classified into four categories depending upon the data being entered or retrieved.

4.10.1. SISO (serial-in, serial-out) Shift Register:

In serial-in, serial-out shift register, data input is in serial form and common clock pulse is applied to each of the flip-flop. After each clock pulse, data moves by one position. The output can be obtained in serial form, as shown in figure below:



Fig. 4.23. SISO Shift Register

- It is the slowest shift register among all the shift registers.
- To store n-bits in a n-bit SISO register, then the minimum “n” clock pulses are required.
- To retrieve n-bits from a n-bit SISO register, then the minimum “(n-1)” clock pulses are required.

4.10.2. SIPO (serial-in, parallel-out) Shift Register

In serial-in, parallel-out shift register, data is applied at the input of register in serial form and the output can be obtained in parallel form after the completely shifting of data in register. Figure below shows the serial input data, and then parallel output.

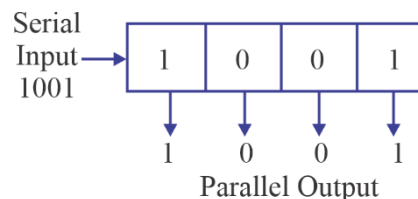


Fig. 4.24. SIPO Shift Register

- To store n-bits in a n-bit SIPO register, the minimum “n” clock pulses are required.
- To retrieve n-bits from a n-bit SIPO register, there is no pulse required.

4.10.3. PISO (parallel-in, serial out) Shift Register

In parallel-in, serial-out shift register, data is loaded into shift register in parallel form and the data output obtained will be serial form as shown in figure below:

- To store n-bit in a n-bit PISO register, a single clock pulse is required.
- To retrieve n-bit from n-bit PISO register, the minimum “(n-1)” clock pulses are required.

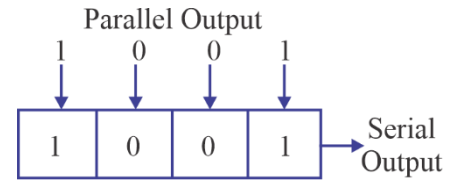


Fig. 4.25. PISO shift register

4.10.4. PIPO (parallel in, parallel out) Shift Register

In parallel-in, parallel-out shift register, data is loaded in parallel form and the data output obtained will be in parallel, as shown in figure below:

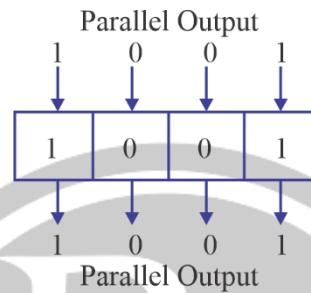


Fig. 4.26. PIPO Shift Register

- To store n-bit in n-bit PIPO register, only a single clock pulse is required.
- To retrieve n-bits from n-bit PIPO register, no clock pulse is required.

Serial Input: The data in the serial form is applied at the serial input after clearing the flip-flops using CLR.

The waveform of serial input shift register is shown below:

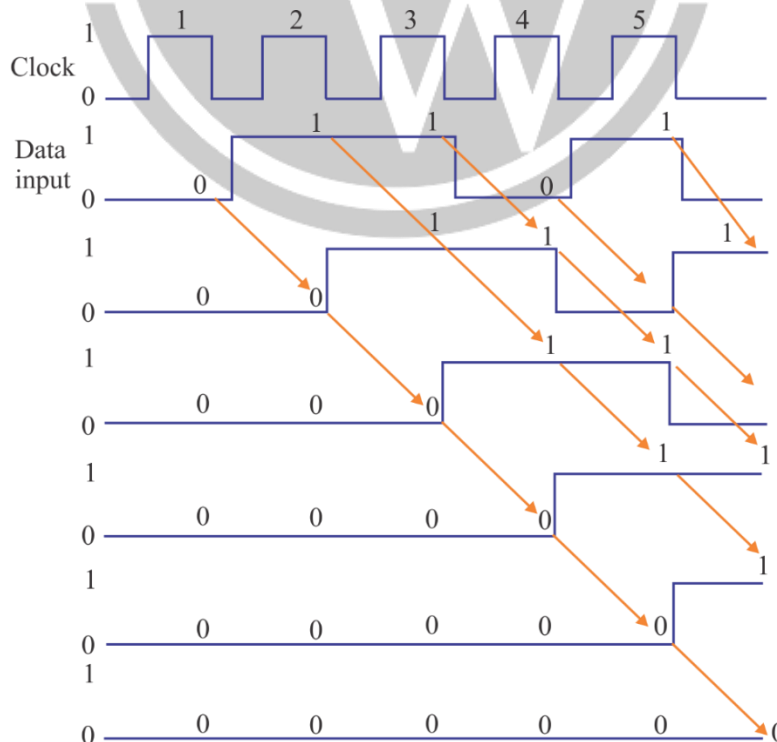


Fig. 4.27.

Parallel Input: Data can be entered in the parallel form making use of the pre-set inputs. Then after clearing the flip-flops, if the data lines are connected to the parallel lines and '1' is applied to the PRESET input.

4.10.5. Universal Shift Register

If the flip-flop outputs of a shift register are accessible, then information entered serially by shifting can be taking out in parallel from the outputs of the flip-flops. If a parallel

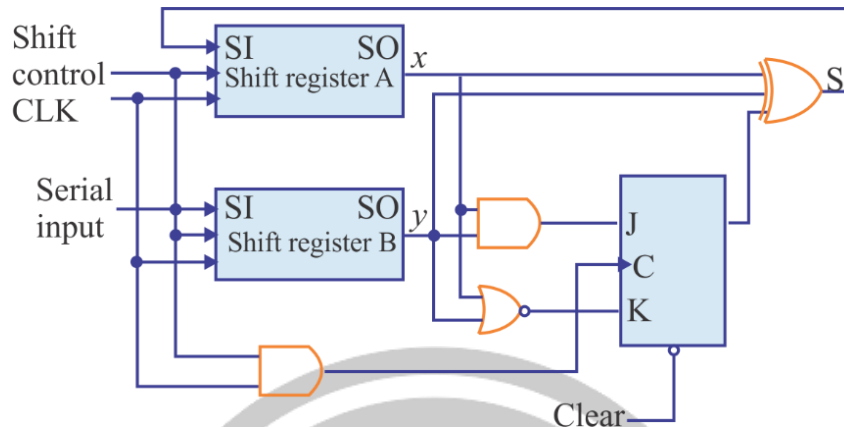


Fig. 4.28. Second form of serial adder

load capability is added to a shift register, then data entered in parallel can be taken out in serial fashion by shifting the data stored in the register. Some shift registers provide the necessary input and output terminals for parallel transfer. They may also have both shift-right and shift-left capabilities.

The most general shift register has the following capabilities:

1. A clear control to clear the register to 0.
2. A clock input to synchronize the operations.
3. A shift-right control to enable the shift-right operation and the serial input and output lines associated with the shift right.
4. A shift-left control to enable the shift-left operation and the serial input and output lines associated with the shift left.
5. A parallel-load control to enable a parallel transfer and the n input lines associated with the parallel transfer.
6. "n" parallel output lines.
7. A control state that leaves the information in the register unchanged in response to the clock.

Other shift registers may have only some of the preceding functions, with at least one. shift operation. A register capable of shifting in one direction only is a unidirectional shift register. One that can shift in both directions is a bidirectional shift register. If the register can shift in both directions and has parallel-load capabilities, it is referred to as a universal shift register. The block diagram symbol and the circuit diagram of a four-bit universal shift register is shown in figure below:

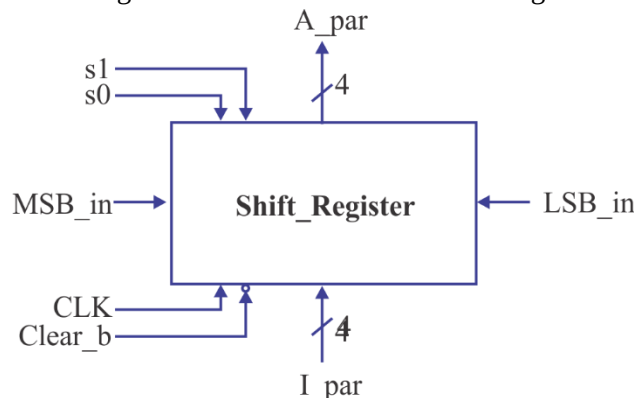


Fig. 4.29. 4-bit Universal Shift Register

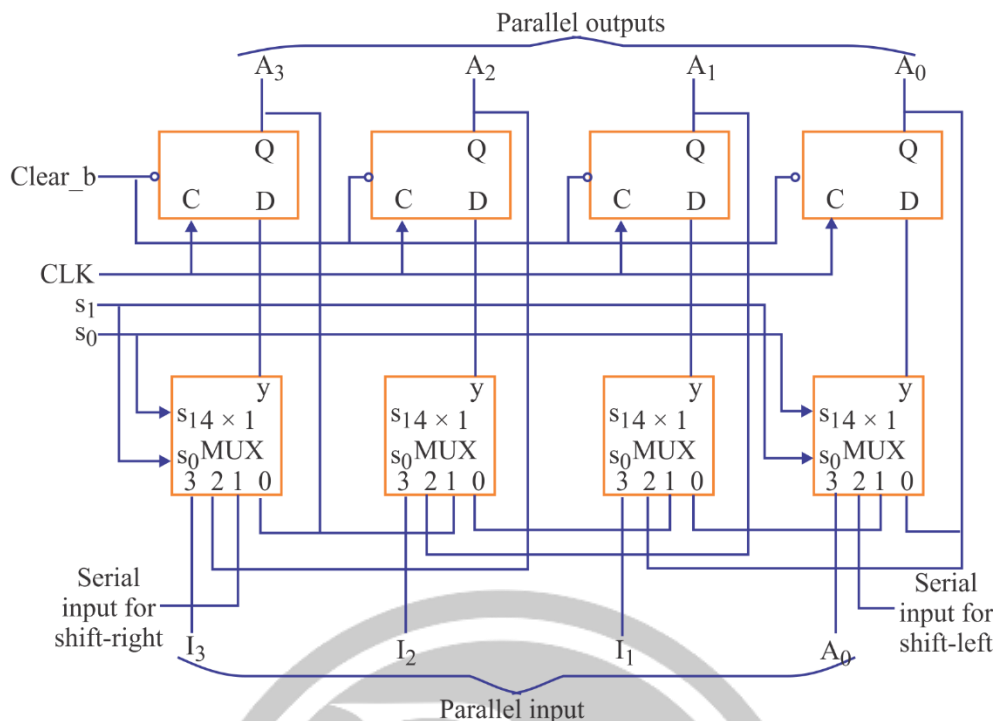


Fig. 4.30. Logic Diagram of 4-bit Universal shift register

The function for the Universal Shift Register is as follows:

Mode Control		Register Operation
S ₁	S ₀	
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	Parallel load

Shift registers are often used to interface digital systems situated remotely from each other. For example, suppose it is necessary to transmit an n-bit quantity between two points. If the distance is far, it will be expensive to use n lines to transmit the its bits in parallel. It is more economical to use a single line and transmit the information serially, one bit at a time. The transmitter accepts the n-bit data in parallel into a shift register and then transmits the data serially along the common line. The receiver accepts the data serially into a shift register. When all n bits are received, they can be taken from the outputs of the register in parallel. Thus, the transmitter performs a parallel-to-serial conversion of data and the receiver does a serial-to-parallel conversion.

4.10.6. Applications of Shift Registers

- (a) **Delay line:** A shift register can be used to introduce a delay (Δt) in signals

$$\Delta t = N \times \frac{1}{f_c}$$

Where N is number of stages & f_c is the clock frequency.

- (b) Serial-to-parallel converter
(c) Parallel-to-serial converter